SOP-AHU Duct Leak Test & Test Report

[CONFIDENTIAL]

[UTTHUNGA]

[IndiQube, South Summit, South End Rd, Vijayarangam Layout, Bengaluru, Karnataka 560004]

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CHANGE HISTORY

This section describes the details of changes that have resulted in the current document.

SL. No.	Document Version No.	Change Description	Reason For change	Date Prepared By	Date Reviewed By	Date Approved By
1	Issue C, V1.0	Initial Draft		Siva Prasad P 22-Sep-25	Mahesh Kumar A 24-Sep-25	Narayana Rao B J 26-Sep-25

1 Purpose

The purpose of this test is to evaluate duct leakage by means of pressure test in Supply Air Duct and Return Air Ducts.

2 Scope

To identify leak through seams or joints by conducted pressure test and sealing of the leaks identified

3 Abbreviations

a. SMACNA – Sheet Metal and Air Conditioning Contractors' National Association

b. CFM – Cubic Feet per Minute (unit of airflow)

c. W.G. – Water Gauge (used for measuring pressure in inches of water column)

4 Terms and Definitions

a. **Duct Leakage Test**: A procedure carried out to determine the amount of air leakage in ductwork by pressurizing the duct system and measuring airflow loss compared to permissible limits.

b. **Permissible Leakage**: The maximum allowable air leakage from a duct system, specified as per SMACNA guidelines or project-specific requirements.

c. **Magnehelic Gauge:** An instrument used to measure differential pressure in ducts, typically in inches of water gauge (W.G.), during the leak testing process.

d. **Leakage Class:** A classification defined by SMACNA that specifies the allowable leakage rate of ducts, expressed as a function of duct surface area and test pressure.

5 Responsibilities

a. Project Supervisors/Technicians: To follow this procedure to carry out duct pressure test.

b. Project Manager/In charge: Responsible for compliance of this procedure.

c. Engineer-QA:

- To verify the test during execution.
- To record the test results and prepare leak test report.

d. Client Representative: To witness and approve the test.

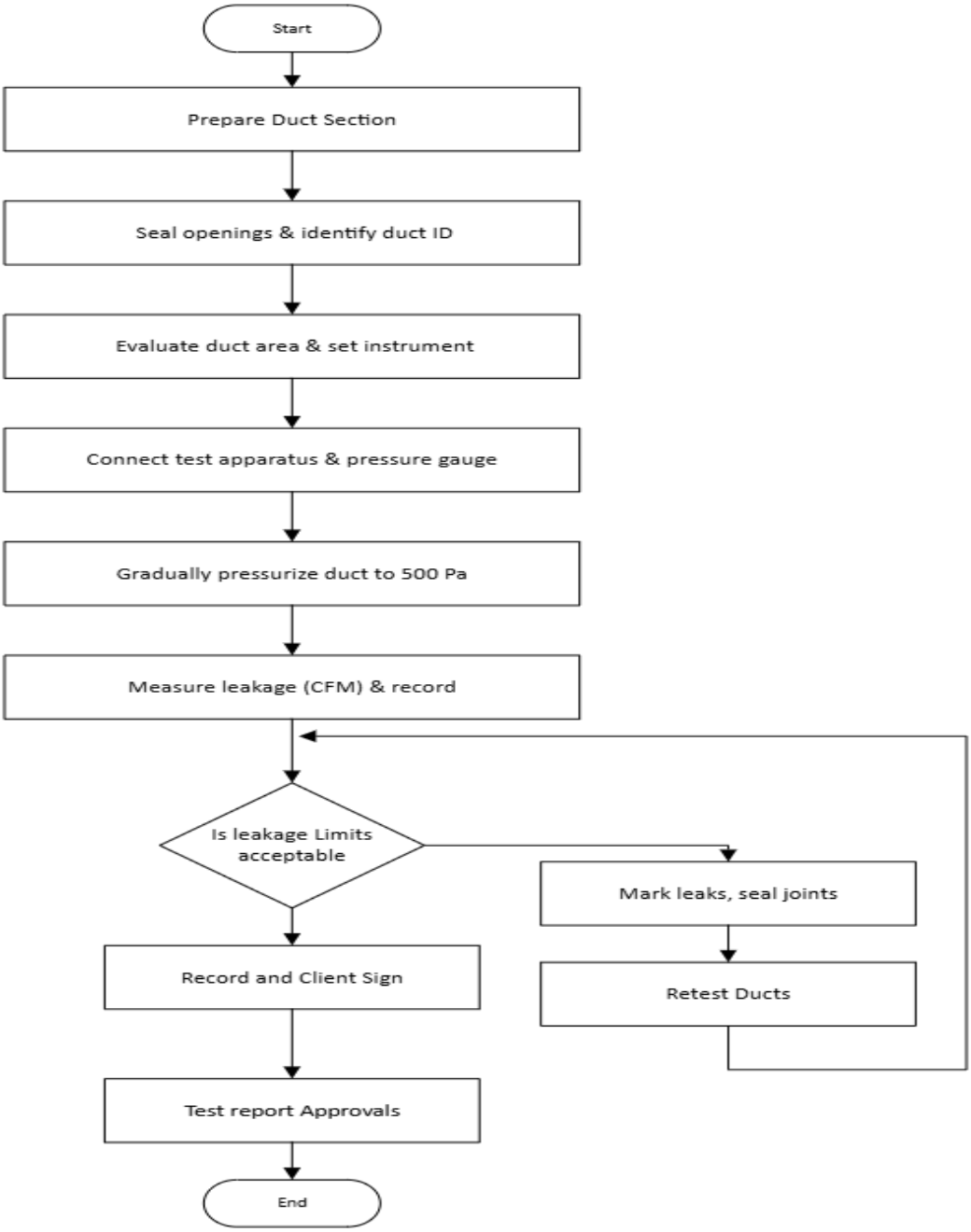
6 RACI

Process	Responsible	Accountable	Consulted	Informed
Preparation of duct section for testing	Site Engineer/ Installer	Project Manager	QA/QC Engineer	Client Representative
Conducting duct leakage test	QA/QC Engineer	Project Manager	Site Engineer	Client Representative
Recording and reviewing test results	QA/QC Engineer	QA/QC Engineer	Site Engineer	Client Representative
Corrective actions for leakage (repairs & sealing)	installer / Site Engineer	Project Manager	QA/QC Engineer	Client Representative
Final approval of test results	QA/QC Engineer	Project Manager	Client Representative	All Stakeholders

7 SIPOC

Supplier	Input	Process and Control		Output	Customer
		Process	Control		
Project Team / Contractor	Ductwork installed, Duct drawings, Test equipment	Duct Leakage Test (as per SMACNA Guidelines)	Pressure gauge, Magnehelic gauge, Test apparatus, QA/QC Procedures	Duct Leak Test Report (PASS/FAIL)	Client Representative / Owner
QA/QC Engineer	Test formats, Acceptance criteria	Seal and pressurize duct, Measure leakage, Record results	Test pressure (500 Pa), Permissible leakage as per SMACNA	Approved Test Record	Project Manager
Site Engineer / Installer	Assembled ducts, Sealing materials	Repairs & retesting in case of failure	Sealants, Gaskets, Retest procedure	Corrected & Retested ducts	QA/QC Manager / Client

8 Overall Process Flow Chart



9 Process Description

9.1 Initial checks:

a. Verify that an adequate and matched electric power source is available for that apparatus.

b. Do not over pressurize ducts. Provide pressure control or pressure relief if test apparatus behavior is unfamiliar, e.g., start test apparatus with flow restricted and gradually build up pressure.

c. Prepare carefully when testing in cold weather. Low pressure influences the effectiveness of sealants and gaskets.

d. Instruct installers to use special care when assembling ducts that will be relatively inaccessible for repair.

e. Conduct required tests before external insulation is applied and before ducts are concealed by building enclosure.

9.2 Test Procedure:

a. Close off and seal openings in the duct section to be tested.

b. Specify the identification numbers to the ducts which is under test in ducting layout drawing.

c. Evaluate the Duct Area (Ft^2 or m^2) with respect to length of the duct.

d. Feed the duct Area and Leakage Class in instrument setting. Based on the Area of the duct, instrument evaluates the permissible leakage as per SMACNA Guideline.

e. Connect the test apparatus to the duct by means of a section of flexible duct.

f. Provide a Magnehelic Gauge or inclined manometer of pressure range 0 inch to 10 inches W.G. on the duct side of shut off valve.

g. To prevent over-pressurizing the ducts, start the blower with variable inlet damper closed. Gradually open the control damper until the duct pressure reaches required pressure.

h. When the pressure of the duct reaches 500 Pa, close the shut off valve.

i. Record the duct leakage quantity result shown in the display and compare the result with respect to permissible leakage. If obtained air leakage is within the permissible limit, record the result and show the result to Client representative for approval.

j. If the test fails to meet the allowable leak rate, survey joints & seams for leaks. Mark each leak and seal all marked joints after shutting down the blower. Do not apply a retest until sealants have set.

k. Take necessary repairs and retest until satisfactory results are obtained.

l. Record the test results in following format.

DUCT LEAK TEST DETAILS			
Client Name	XXXXX	Test Report	
Project Name	<Clean Room-3>	Test Date	
Duct ID Number		Duct Type	<Supply/Return>
Drawing Number		AHU Number	

m. Duct shall be identified in terms of area classified and supply or return duct.

n. Identify the ducts in the drawing (which are subjected to leak test) and copy of relevant portion of drawing shall be enclosed.

o. Record test results in the following format, paste test result print out in take approval from Client Representative.

DUCT LEAK TEST REPORT				
Duct Identification No.	Duct Area, Ft^2	Permissible leakage, CFM	Measured Leakage, CFM	Result PASS/FAIL

9.3 Acceptance Criteria:

Duct work Air Leakage percentage should be within permissible limit as per SMACNA Guideline or agreed to between the owner/buyer and Utthunga Service team.

Prepared by:	Reviewed by:	Approved & authorized by:
Sign & Date: Name: Designation:	Sign & Date: Name: Designation:	Sign & Date: Name: Designation:

10 Information Security requirements

a. Procedures shall be implemented for the management of removable media in accordance with the classification scheme adopted by the organization.

b. Media containing information shall be protected against unauthorized access, misuse, or corruption during transportation.

c. Development, testing, and operational environments shall be separated to reduce the risks of unauthorized access or changes to the operational environment.

d. The information security related requirements shall be included in the requirements for new information systems or enhancements to existing information systems.

e. Information involved in application services passing over public networks shall be protected from fraudulent activity, contract dispute and unauthorized disclosure and modification.

11 References

a. Li SMACNA Guideline for Duct leak test

b. Duct Leak Factor (F) is calculated by the equation, F=CL x (P)0.65

Where:

F= Leakage Factor (CFM/100 Ft2 of Duct Surface Area)

CL= Leakage Class

P= Static Pressure in the Duct (mmWC)

c. For Leakage Classification of Duct, refer SMACNA Guideline.

12 Metrics

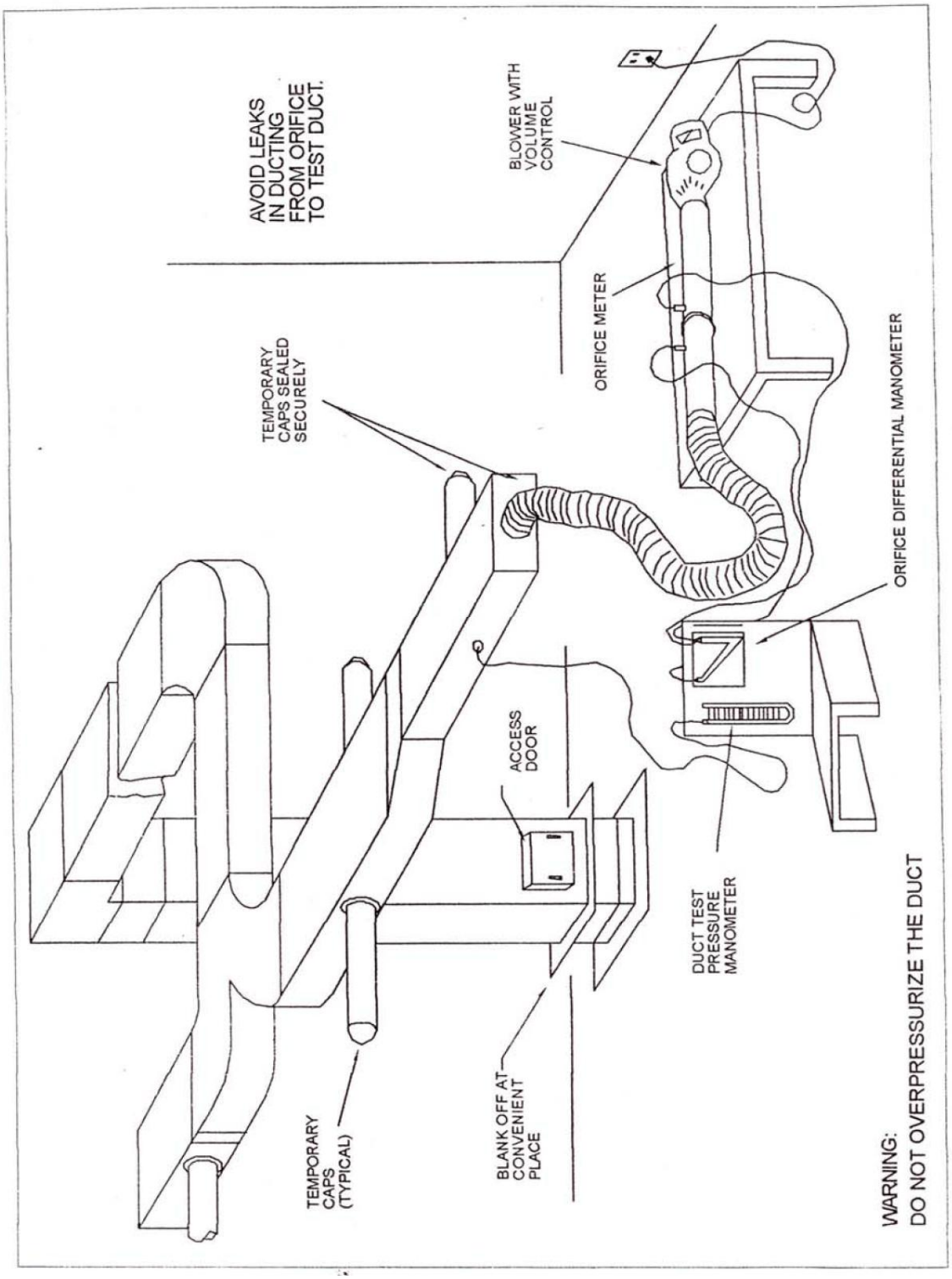
a. Duct Area (Ft^2/m^2)

b. Test Pressure (e.g., 500 Pa)

c. Measured vs. Permissible Leakage (CFM) as per SMACNA standards

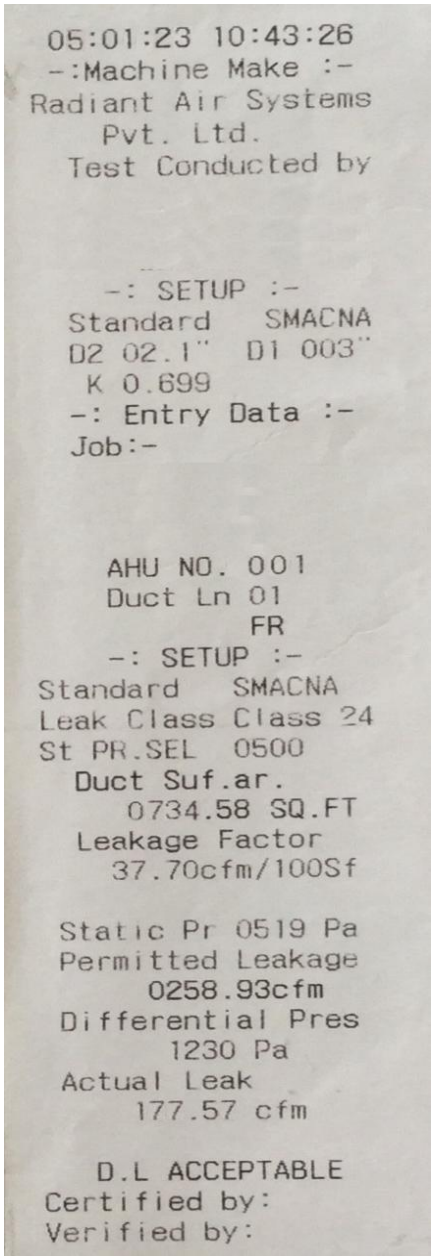
13 Annexures

13.1 Annexure-A: PROCEDURE


AVOID LEAKS IN DUCTING FROM ORIFICE TO TEST DUCT.
BLOWER WITH VOLUME CONTROL
ORIFICE METER
TEMPORARY CAPS SEALED SECURELY
ORIFICE DIFFERENTIAL MANOMETER
ACCESS DOOR
DUCT TEST PRESSURE MANOMETER
TEMPORARY CAPS (TYPICAL)
BLANK OFF AT CONVENIENT PLACE
WARNING:
DO NOT OVERPRESSURIZE THE DUCT

DUCT LEAK TEST DETAILS	
Client Name	**XXXXX India Pvt. Ltd.**
Project Name	**‹Clean Room-3›**
Test Report No.	
Test Date	
AHU No	**AHU-01**
Duct Type	**Supply**
Drawing Ref. No.	
TEST RESULTS	
Duct ID No.	**Line-01**
Duct Area in Ft^2	**734.58**
Test Pressure in Pa	**1230**
Permissible Air Leakage in CFM	**258.93**
Measured Leakage in CFM	**177.57**
Result	**PASS**

DUCT LEAK TEST RAW DATA



05:01:23 10:43:26
-:Machine Make :-
Radiant Air Systems
Pvt. Ltd.
Test Conducted by

-: SETUP :-
Standard SMACNA
D2 02.1" D1 003"
K 0.699
-: Entry Data :-
Job:-

AHU NO. 001
Duct Ln 01
FR
-: SETUP :-
Standard SMACNA
Leak Class Class 24
St PR.SEL 0500
Duct Suf.ar.
0734.58 SQ.FT
Leakage Factor
37.70cfm/100Sf

Static Pr 0519 Pa
Permitted Leakage
0258.93cfm
Differential Pres
1230 Pa
Actual Leak
177.57 cfm

D.L ACCEPTABLE
Certified by:
Verified by:

Remarks:

DUCT LEAK TEST DETAILS		DUCT LEAK TEST RAW DATA
Client Name	**XXXXX India Pvt. Ltd.**	
Project Name	**‹Clean Room-3›**	
Test Report No.		
Test Date		
AHU No	**AHU-01**	
Duct Type	**Return**	
Drawing Ref. No.		
TEST RESULTS		
Duct ID No.	**Line-02**	
Duct Area in Ft^2	**618.21**	
Test Pressure in Pa	**1230**	
Permissible Air Leakage in CFM	**258.93**	
Measured Leakage in CFM	**146.56**	
Result **PASS/FAIL**	**PASS**	

05:01:23 11:43:26
-:Machine Make :-
Radiant Air Systems
Pvt. Ltd.
Test Conducted by

-: SETUP :-
Standard SMACNA
D2 02.1" D1 003"
K 0.699
-: Entry Data :-
Job:-

3 FL.HOSUR

AHU NO. 001
Duct Ln 02
FR
-: SETUP :-
Standard SMACNA
Leak Class Class 24
St PR.SEL 0500
Duct Suf.ar.
0618.21 SQ.FT
Leakage Factor
37.70cfm/100Sf

Static Pr 0519 Pa
Permitted Leakage
0258.93cfm
Differential Pres
1230 Pa
Actual Leak
149.56 cfm

D.L ACCEPTABLE
Certified by:
Verified by:

Remarks:

14 Forms

a. Duct Work Leak Test Report UTEPCTMP11